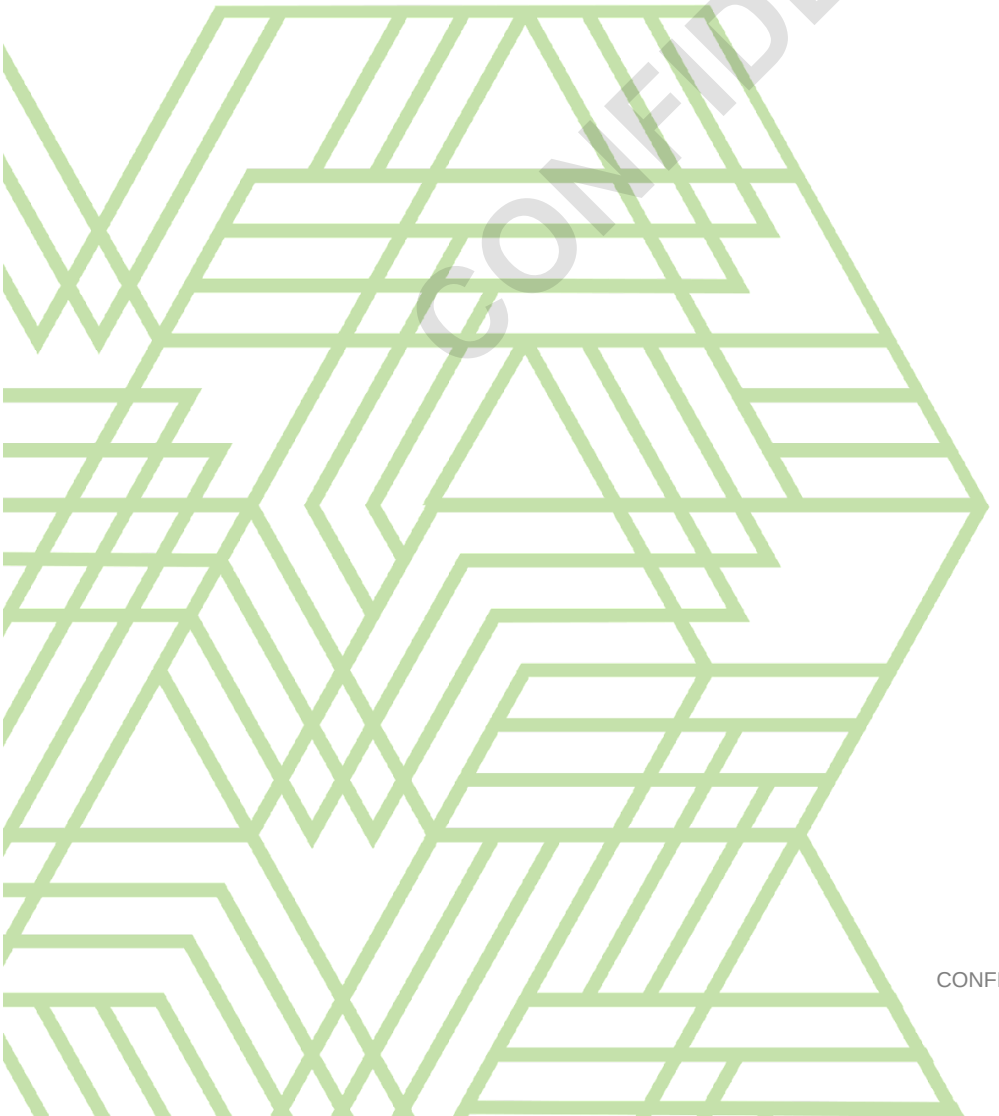
ATM33/e General Purpose ADC

Application Note

SUMMARY: This document describes the General Purpose ADC (Analog to Digital Converter) hardware operation of the ATM33/ATM33e Wireless SoC Series.

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Acronyms and Abbreviations

Acronyms	Definition
ATM33	ATM3325 ATM3330
ATM33e	ATM3330e
ATM33/e	ATM33/ATM33e
GADC	General purpose Analog to Digital Converter
SDK	Software Development Kit
SoC	System-on-Chip

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1. Overview

This document describes the operation of the GADC driver for the ATM33/e Wireless SoC Series.

See Table 1 for the applicable SoCs and packages:

SoC	SoC Part Number	EVK Part Number	Package
ATM33 Series	ATM3325-5DCAQK	ATMEVK-3325-QK	40-pin 5x5 mm QFN package
	ATM3325-5LCAQK	ATMEVK-3325-LQK	40-pin 5x5 mm QFN with extended storage package
	ATM3330-5DCAQN	ATMEVK-3330-QN	56-pin 7x7 mm QFN package
ATM33e Series	ATM3330E-5DCAQN	ATMEVK-3330e-QN	56-pin 7x7 mm QFN package

Table 1 - Applicable SoCs and Packages

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2. GADC Driver Capabilities

- Supports up to 9 physical channels (5 fixed and 4 general-purpose)
  - Fixed: VBAT, VSTORE, VBATLI, AVDD1 (CORE), temperature
  - General-purpose: P4, P5, P6, and P7 can be individually used as single inputs, or together as P4/P5 and P6/P7 to be differential inputs
  - As shown in Table 2, there are 11 programmable channels
    - 2 additional programmable channels for differential configuration
- Adjustable gain for each channel
- 11 bit resolution at 2 Million Samples/second

2.1 Channel and Gain Selection

The available GADC programmable channels and corresponding information are summarized in Table 2

Channel number	Channel name	Port	Gain	Resolution (mV)	Voltage Range	
	(DGADC_CH)		(DGADC_EXT)		min	max
1	VBATT	VBAT/LI (fixed)	GAIN_EXT_QUARTER	3.91	0	3.3
			GAIN_EXT_HALF	1.95	0	1.6
2	VSTORE	VSTORE (fixed)	GAIN_EXT_QUARTER	3.91	0	3.2
			GAIN_EXT_HALF	1.95	0	1.6
3	CORE	AVDD1 (fixed)	GAIN_EXT_HALF	1.95	0	1.5
4	TEMP	temperature (fixed)	GAIN_EXT_X1	0.40 °C/LSB	-40 °C	85 °C
5	PORT1_DIFFERENTIAL	P4/P5	GAIN_EXT_QUARTER	3.91	-3	3
			GAIN_EXT_HALF	1.95	-1.5	1.5
			GAIN_EXT_X1	9.77	-0.75	0.75
			GAIN_EXT_X2	4.89	-0.375	0.375
6	PORT0_DIFFERENTIAL	P6*/P7**	GAIN_EXT_QUARTER	3.91	-3	3
			GAIN_EXT_HALF	1.95	-1.5	1.5

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Channel number	Channel name	Port	Gain	Resolution (mV)	Voltage Range	
	(DGADC_CH)		(DGADC_EXT)		min	max
			GAIN_EXT_X1	9.77	-0.75	0.75
			GAIN_EXT_X2	4.89	-0.375	0.375
7	PORT1_SINGLE_ENDED_0	P4	GAIN_EXT_QUARTER	3.91	0	3
			GAIN_EXT_HALF	1.95	0	1.5
8	PORT1_SINGLE_ENDED_1	P5	GAIN_EXT_QUARTER	3.91	0	3
			GAIN_EXT_HALF	1.95	0	1.5
9	PORT0_SINGLE_ENDED_0	P6*	GAIN_EXT_QUARTER	3.91	0	3
			GAIN_EXT_HALF	1.95	0	1.5
10	PORT0_SINGLE_ENDED_1	P7**	GAIN_EXT_QUARTER	3.91	0	3
			GAIN_EXT_HALF	1.95	0	1.5
11	LI_ION_BATT	VBAT/LI (fixed)	GAIN_EXT_X1	5.86	0	4.2

*** Not supported in 5x5 mm with extended storage package (ATM3325-5LCAQK)**
****Not supported in 5x5 mm with harvesting package(ATM3330E-5DCAQK)**

Table 2 - Basic GADC Channel Information

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Fixed channels VBAT, VSTORE, VBATLI, and AVDD1 can be measured on the EVB through the labeled pins in Figure 1. General-purpose channels will require voltage input to GPIOs P4, P5, P6, or P7.

GAIN_EXT_QUARTER, GAIN_EXT_HALF, GAIN_EXT_X1, and GAIN_EXT_X2 are internal gain settings. These gain settings are represented by the voltage range in the last two columns of Table 2.

Note that the selected gain for a channel affects the voltage range that can be measured. If the maximum of a gain setting is surpassed, the GADC result will rail at the max voltage. It is advised to choose the largest gain setting possible for the voltage for the best resolution.

2.2 GADC Output Path

The GADC software driver takes raw output code of the GADC and converts it into a voltage value using a calibrated slope and offset. Each channel has its own slope and gain values, and the driver scales the GADC code according to the gain setting.

3. Running `GADC_test`

The sample SDK application GADC_test demonstrates the basic functionality of the GADC. It takes 100 samples of the selected channel in 100 ms intervals and prints the resulting samples in the debug console.

These notes will show an example of measuring the single-ended external channel 7. This channel reads the input voltage that is applied to GPIO P4.

3.1 Getting Started

The sample GADC_test can be found in the SDK under (atmosic_sdk/platform/atm33/ATM33xx-5/examples). Refer to the README file in the GADC_test folder and the Reference Documents section for related documents.

Before running the test, the channel and gain should be specified by modifying `CFLAGS -DGADC_CH` and `-DGADC_EXT` in the makefile located in the GADC_test folder. The

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configuration flag values supported by the driver can be seen in Table 2 under the `Channel name` and `Gain` columns.

The channel has been set to `PORT1_SINGLE_ENDED_0` and gain to `GAIN_EXT_QUARTER` in the following makefile:

```
################################################################################
#
## @file makefile
#
## @brief GADC test
#
## Copyright (C) Atmosic 2021-2023
#
################################################################################

include ../../user/common.mk

DEBUG := 1
DRIVERS := interrupt gadc timer no_ble
UU_TEST := GADC_test gadc

CFG_GCC_PRINT_FLOAT := 1

CFLAGS += -DGADC_CH=PORT1_SINGLE_ENDED_0 DGADC_EXT=GAIN_EXT_QUARTER

ifdef GADC_12_BIT
CFLAGS += -DENABLE_GADC_12_BIT
endif

include $(COMMON_USER_DIR)/app.mk
```

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3.2 Board Configuration

Refer to Table 2 to determine the port to apply the voltage that will be measured. To apply external voltages to internally generated channels (VBAT, AVDD1, and VBATLI), follow the subsequent sections for specific jumper configurations.

3.2.1 VBAT

If externally supplying VBAT, make sure that the J28 jumper is connecting VBAT and VBATLI (two left pins). There should be no jumpers on J21 (middle pin VBAT(LI) should not be connected to any other pins). The VBAT voltage should be applied to the VBAT/LI test loop.

3.2.2 AVDD1

AVDD1 must be tied to AVDD1P, no changes are needed on the board for this measurement.

3.2.3 VBATLI

If externally supplying VBATLI, remove the J28 jumper and leave all three pins unconnected. There should be no jumpers on J21 (middle pin VBAT(LI) should not be connected to any other pins). The VBATLI voltage should be applied to the VBAT/LI test loop.

For the fixed channels, the GADC will select the desired channel as the input.

For general-purpose channels, refer to the two rows of GPIOs in the board layout configuration below. The channel selected will determine which of the GPIOs input voltage should be applied. It is recommended that all other external inputs besides the measured channel be grounded to minimize interference.

Note: *For general-purpose channels, there is a capacitive input impedance of 400 fF.*

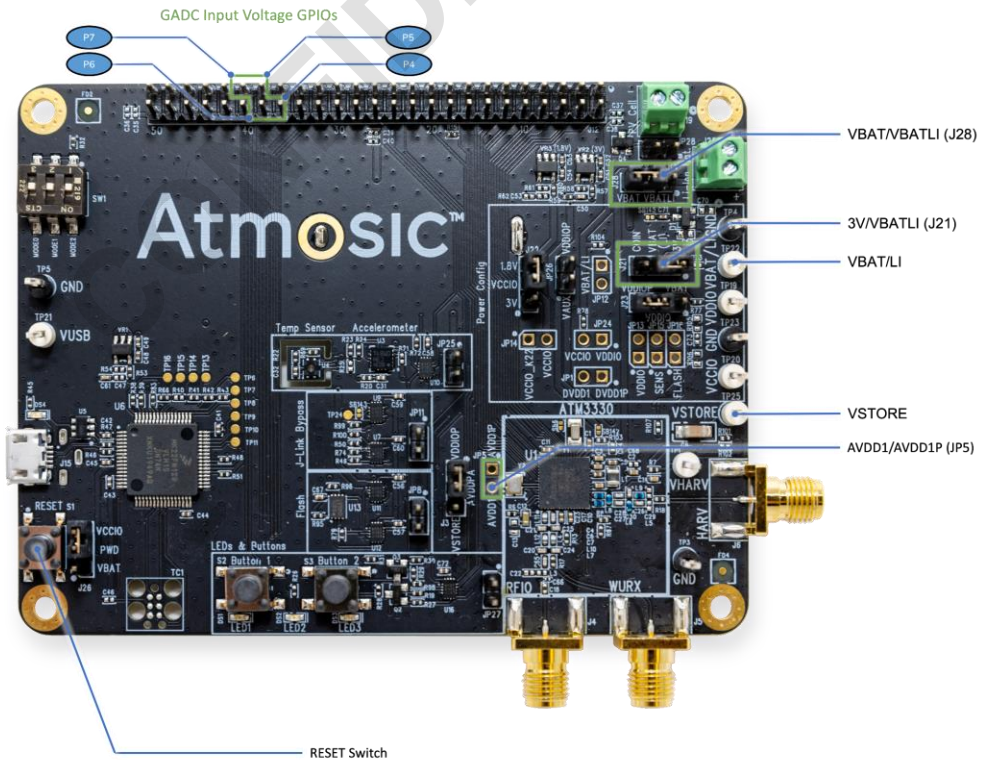


Figure 1 - ATM33/e v3.x EVB Configuration with Supported GADC GPIOs

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3.3 Building, Programming, and Running GADC_test

To build, program, and run in a single command, go to the GADC_test folder, and use the command:

```
make run_all BOARD=ATMEVK_3330_QN
```

Note: *The BOARD parameter will be different for ATM3325 and ATM3330e.*

100 samples will print out in the device's COM port terminal. To rerun GADC_test, press the RESET button located at the bottom left corner on the EVB.

The `Result` is the voltage value after applying the calibrated offset and gain to the `Raw` GADC output value. The results below are with the configuration flags `-DGADC_CH=PORT1_SINGLE_ENDED_0` and `-DGADC_EXT=GAIN_EXT_QUARTER` with an input voltage of 1.0 V applied to GPIO P4.

1.0 V input for PORT1_SINGLE_ENDED_0 and GAIN_EXT_QUARTER:

```
Result: 1.000, Raw: 124.5
Result: 1.000, Raw: 124.5
Result: 1.000, Raw: 124.5
Result: 1.000, Raw: 124.5
```

If the gain is changed to `-DGADC_EXT=GAIN_EXT_HALF`, the GADC output `Raw` value is twice as large compared to `GAIN_EXT_QUARTER`. Note that `Result` is still the same due to the driver scaling to the correct gain setting.

1.0 V input for PORT1_SINGLE_ENDED_0 and GAIN_EXT_HALF:

```
Result: 0.999, Raw: 249.5
Result: 0.999, Raw: 249.5
Result: 0.999, Raw: 249.5
```

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4. Evaluating the GADC

4.1 Performance

Channel	Typical Variation	Unit
Temperature	17 (-40ºC to 85ºC)	ºC
	9 (0ºC to 50ºC)	ºC
Single Input (single-ended or fixed)	10	mV
Differential Input	20	mV
VBATLI	35	mV

Reference Documents

Title	Document Number
SDK User Guide	ATM-QSGSDK

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Revision History

Date	Version	Description
May 17, 2023	0.50	Initial version created.

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